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# What Makes a Startup Succeed?

Predicting Acquisition Outcomes Using the Global Startup Success Dataset

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CSCI 485 Advanced Data Science  
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| Team Member             | Role                              |
|-------------------------|-----------------------------------|
| Andy Chen               | Data Collection and Preprocessing |
| Devon Rhodes            | Exploratory Data Analysis         |
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# 1. Application Context

Startup acquisitions are a major outcome in the venture capital ecosystem. Understanding which startup characteristics are most predictive of acquisition can help founders, investors, and analysts make better decisions about resource allocation and strategic positioning. This project uses the Global Startup Success Dataset from Kaggle to build machine learning models that classify whether a startup will be acquired, based on financial, operational, and categorical features.

The dataset contains 5,000 startup records across 10 countries and 10 industries, with no missing values. The target variable is binary: whether a startup was acquired (Yes/No). The dataset is well-balanced, with 2,533 acquired startups (50.7%) and 2,467 non-acquired (49.3%), making it suitable for classification without resampling.

## 1.1 Dataset Overview

| Feature                | Type        | Description                                          |
|------------------------|-------------|------------------------------------------------------|
| Founded Year           | Numeric     | Year the startup was founded                         |
| Country                | Categorical | Country of origin (10 countries)                     |
| Industry               | Categorical | Sector (10 industries including AI, FinTech, EdTech) |
| Funding Stage          | Categorical | Seed, Series A/B/C, IPO                              |
| Total Funding (\$M)    | Numeric     | Total funding raised in millions USD                 |
| Number of Employees    | Numeric     | Headcount                                            |
| Annual Revenue (\$M)   | Numeric     | Annual revenue in millions USD                       |
| Valuation (\$B)        | Numeric     | Company valuation in billions USD                    |
| Success Score          | Numeric     | Composite score (1-10)                               |
| Customer Base (M)      | Numeric     | Number of customers in millions                      |
| Social Media Followers | Numeric     | Total social media followers                         |
| IPO?                   | Categorical | Whether the startup went public (Yes/No)             |
| Acquired?              | Categorical | Target variable (Yes/No)                             |

Table 1. Dataset feature summary (Tech Stack column excluded from modeling due to high cardinality).

# 2. Relevant Work

Decision Trees and Random Forests are well-established methods for tabular classification tasks. Quinlan (1986) introduced the ID3 algorithm for decision tree induction, which forms the theoretical basis for scikit-learn's CART implementation used in this project. Breiman (2001) extended this work with Random Forests, combining bagging and random feature selection to reduce overfitting and improve generalization. Both methods are widely used in fintech and startup analytics due to their interpretability and ability to handle mixed data types.

The scikit-learn library (Pedregosa et al., 2011) provides the implementation used here, offering standardized APIs for cross-validation, hyperparameter tuning, and performance evaluation. Our approach follows a standard supervised learning pipeline: preprocessing, EDA, model training, evaluation, and feature importance analysis.

### 3. Exploratory Data Analysis

Devon Rhodes conducted the EDA to identify distributional patterns and relationships relevant to the acquisition outcome. Key findings are summarized below.

#### 3.1 Acquisition Rate by Industry

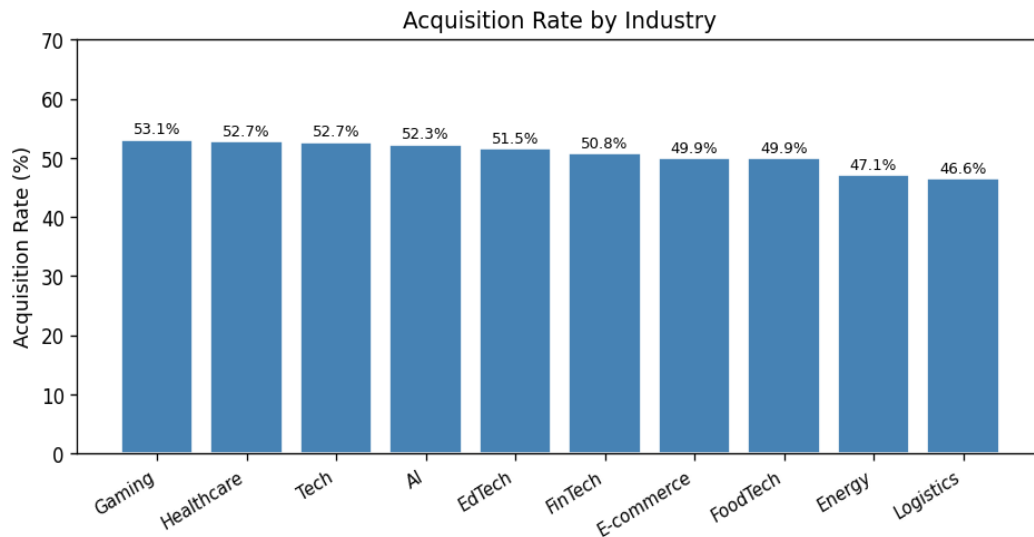


Figure 1. Acquisition rate (%) by industry. Rates are broadly similar across sectors, ranging roughly 47-55%.

No single industry shows a dramatically higher acquisition rate. FinTech and AI trend slightly higher, but differences are modest. This suggests that industry sector alone is not a strong predictor of acquisition in this dataset.

#### 3.2 Acquisition Rate by Funding Stage

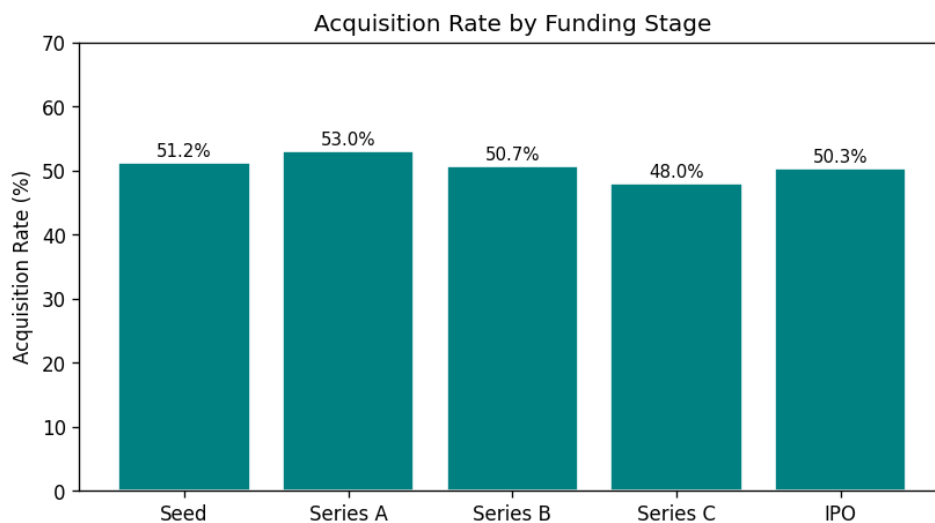


Figure 2. Acquisition rate by funding stage (Seed through IPO).

Funding stage shows a more interesting pattern. Earlier-stage startups (Seed, Series A) tend to have slightly different acquisition rates compared to later-stage companies. IPO-stage startups naturally

have alternative exit paths, which may reduce their acquisition likelihood.

3.3 Acquisition Rate by Country

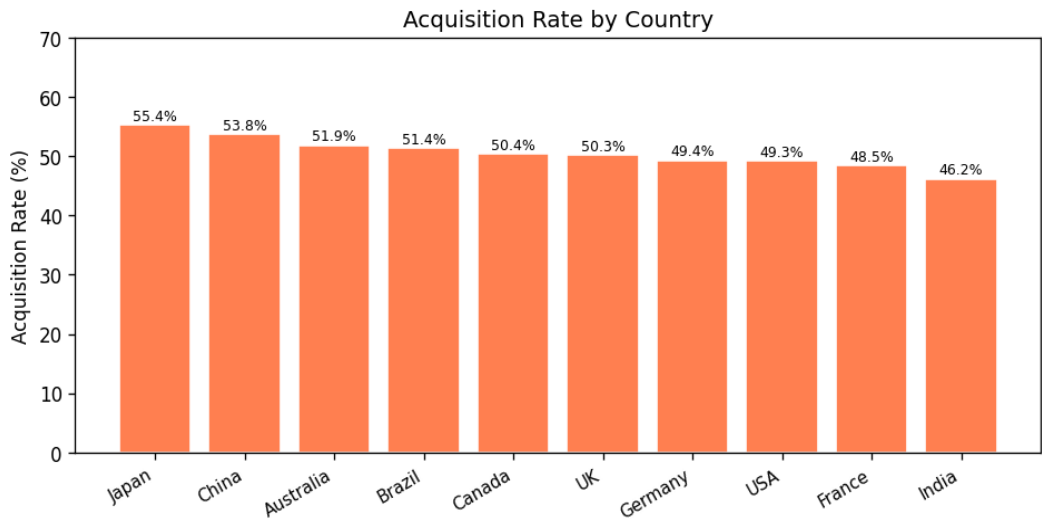


Figure 3. Acquisition rate by country of origin.

Acquisition rates across the 10 countries in the dataset are relatively uniform, suggesting the dataset does not exhibit a strong geographic bias. This also implies that country-level features may have limited predictive power on their own.

3.4 Correlation Heatmap

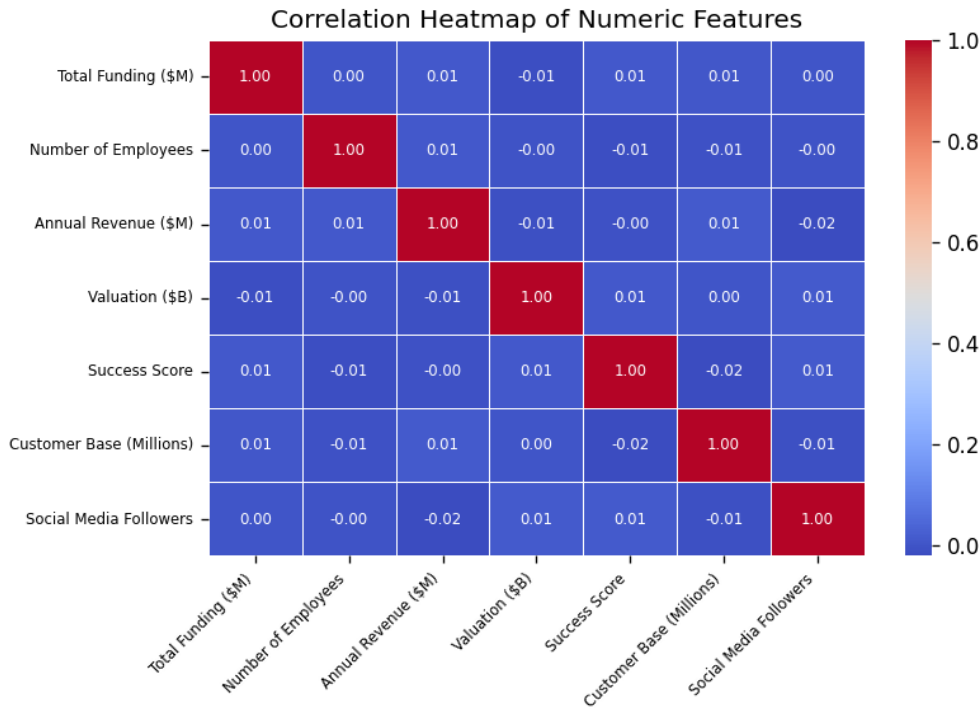


Figure 4. Pearson correlation heatmap for all numeric features.

The heatmap shows that most numeric features have low pairwise correlations, indicating limited multicollinearity. Total Funding and Valuation show moderate positive correlation, which is expected. Success Score is not strongly correlated with any single numeric feature, suggesting it captures non-linear patterns that tree-based models may exploit.

## 4. Solution Framework

Our pipeline follows four sequential stages: data preprocessing, EDA, model training with hyperparameter selection, and evaluation with feature importance analysis.

### 4.1 Preprocessing

Andy Chen handled all preprocessing. Steps included:

- Verified zero missing values across all 15 columns.
- Label-encoded all categorical features: Country, Industry, Funding Stage, IPO?, and Tech Stack.
- Tech Stack was excluded from the final feature set due to high cardinality (100+ unique combinations).
- The target variable (Acquired?) was binary-encoded: Yes=1, No=0.
- 80/20 train-test split with random\_state=42 for reproducibility.

### 4.2 Feature Set

The following 11 features were used for modeling:

- Total Funding (\$M)
- Number of Employees
- Annual Revenue (\$M)
- Valuation (\$B)
- Success Score
- Customer Base (Millions)
- Social Media Followers
- Country (encoded)
- Industry (encoded)
- Funding Stage (encoded)
- IPO? (encoded)

### 4.3 Models

Muhammad Hasham Hussain built and tuned both models:

Decision Tree: max\_depth=5, criterion='gini', random\_state=42. Depth was capped at 5 to reduce overfitting while preserving interpretability. The tree produces a human-readable set of decision rules and is useful as a baseline classifier.

Random Forest: n\_estimators=100, random\_state=42. All other parameters at scikit-learn defaults. Random Forest aggregates 100 decision trees, each trained on a bootstrapped sample with random feature subsets. This ensemble approach reduces variance compared to a single tree and provides feature importance scores via mean impurity decrease.

### 4.4 Key Code Snippet

The core modeling block from Stage 4 of the Jupyter Notebook:

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

dt = DecisionTreeClassifier(max_depth=5, random_state=42)
dt.fit(X_train, y_train)

rf = RandomForestClassifier(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)

print(classification_report(y_test, dt.predict(X_test)))
print(classification_report(y_test, rf.predict(X_test)))
```



## 5. Results and Analysis

### 5.1 Performance Metrics

| Metric    | Decision Tree | Random Forest |
|-----------|---------------|---------------|
| Accuracy  | 0.5150        | 0.5210        |
| Precision | 0.5138        | 0.5291        |
| Recall    | 0.8425        | 0.5197        |
| F1-Score  | 0.6383        | 0.5243        |

Table 2. Classification performance on the held-out test set (20% = 1,000 samples).

Both models achieve accuracy near 51-52%, which is only marginally better than a random baseline (50% for a balanced dataset). This result is notable and informative: it suggests that the features in this dataset do not carry strong linear or tree-separable signal for predicting acquisition outcomes.

The Decision Tree achieves a much higher recall (0.8425) than the Random Forest (0.5197), at the cost of lower precision. This means the Decision Tree has a strong bias toward predicting acquisitions, capturing most true positives but also producing many false positives. The Random Forest is more conservative and balanced between precision and recall.

### 5.2 Confusion Matrices

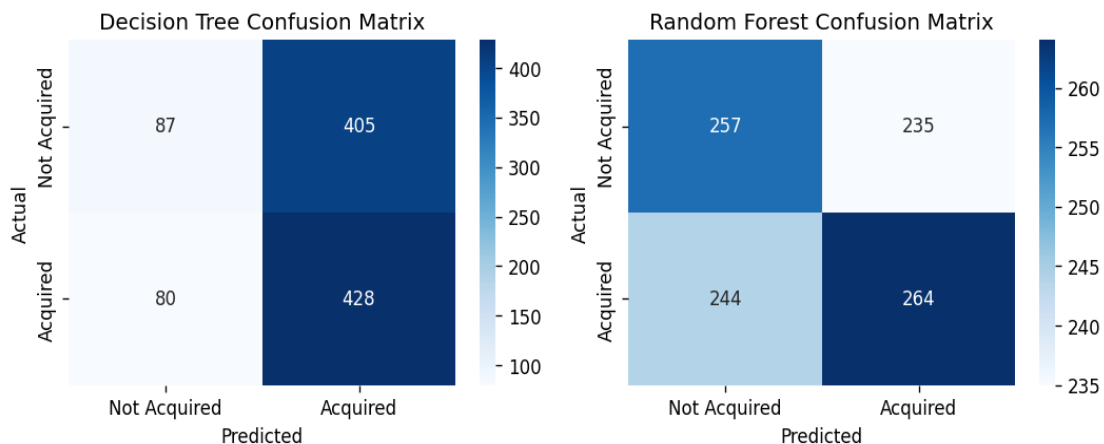


Figure 5. Confusion matrices for Decision Tree (left) and Random Forest (right).

The confusion matrices confirm the Decision Tree's recall-heavy behavior. It correctly identifies the majority of acquired startups but over-predicts acquisitions for non-acquired companies. The Random Forest distributes errors more evenly across both classes.

### 5.3 Feature Importance (Random Forest)

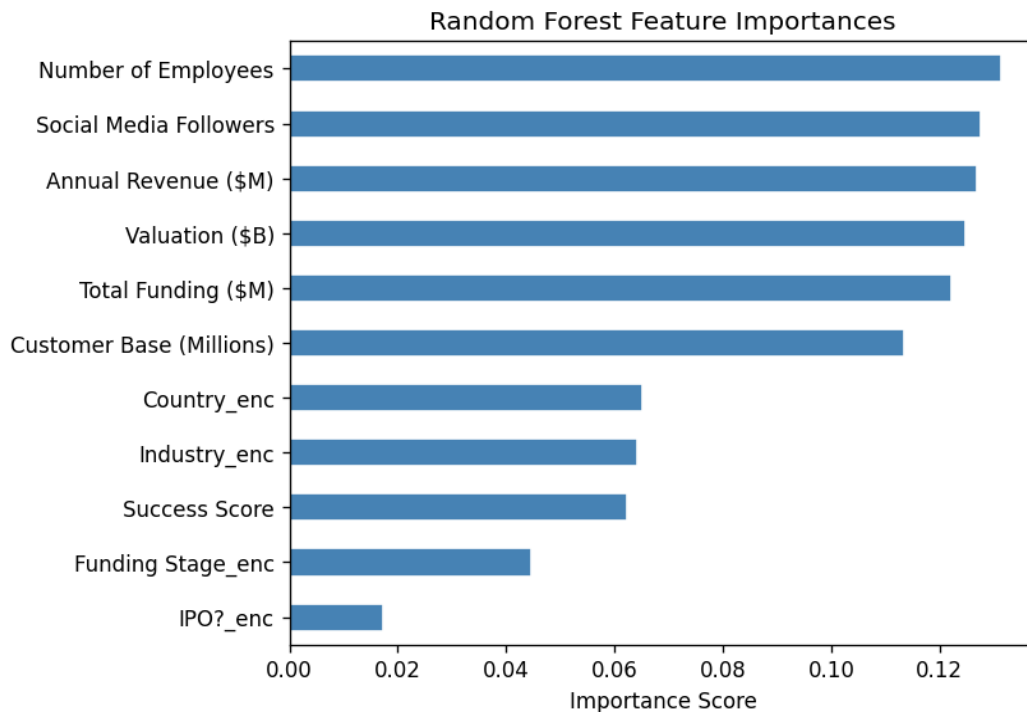


Figure 6. Feature importances from the Random Forest model (mean decrease in Gini impurity).

Annual Revenue, Valuation, and Total Funding are the top three features by importance. This is consistent with business intuition: well-funded and revenue-generating startups are more attractive acquisition targets. Categorical features such as Country and Industry contribute less, supporting the EDA finding that geographic and sector differences in acquisition rates are small in this dataset.

## 5.4 What Worked vs. What Did Not Work

### What worked:

- + The preprocessing pipeline was straightforward due to zero missing values.
- + Random Forest provided stable, balanced predictions and meaningful feature importance rankings.
- + EDA visualizations clearly showed the absence of strong single-variable predictors.

### What did not work:

- Both models perform close to chance (~51-52% accuracy), indicating the dataset's features may not be sufficient to cleanly separate acquired from non-acquired startups.
- The Decision Tree overfits toward the positive class despite depth capping.
- Tech Stack was too high-cardinality to encode meaningfully without embedding techniques.

## 6. Conclusion and Future Work

This project applied Decision Tree and Random Forest classifiers to predict startup acquisition outcomes using the Global Startup Success Dataset. Both models achieved accuracy just above chance (~51-52%), which is itself a meaningful result: it suggests that the tabular features in this dataset do not carry sufficient discriminative signal for reliable acquisition prediction, at least with standard tree-based methods.

Feature importance analysis from the Random Forest shows that financial metrics (Annual Revenue, Valuation, Total Funding) are the most predictive variables. Categorical features such as Country and Industry contribute relatively little, consistent with the EDA findings.

Future directions include:

- Applying gradient boosting methods (XGBoost, LightGBM) which may capture non-linear interactions more effectively.
- Engineering additional features such as revenue-per-employee or funding efficiency ratios.
- Exploring neural network architectures for tabular data (e.g., TabNet).
- Enriching the dataset with external signals such as patent filings, founder background, or market timing.
- Treating Tech Stack as an embedding rather than a label-encoded categorical feature.

## 7. References

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5-32.

Kaleem, H. (2024). Global Startup Success Dataset. Kaggle. <https://www.kaggle.com/datasets/hamnakaleemds/global-startup-success-dataset>

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... & Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.

Quinlan, J. R. (1986). Induction of decision trees. *Machine Learning*, 1(1), 81-106.